Nuclear Decay as a Spectral Bifurcation of the Flag Condensate: Instanton Action, Bogoliubov Mode Mixing, and Geometric Unification

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Abstract

We present a particle-free formulation of nuclear decay and quantum tunneling within the Algebraic Conductance State (ACS) flag-condensate ontology. By representing the nucleus as a confined standing-wave phase lock of canonically conjugate real field components ϕ (cosine mode) and π (sine mode), decay is reinterpreted as a spectral bifurcation into a travelling wave driven by geometric phase slips across the colour-confining throat. We derive the WKB Gamow transmission factor $T = \mathrm{e}^{-2W}$ from the Euclidean deformation of the BPST instanton action $\delta S(R)$ on the curved throat metric. Utilizing a numerically stable 2×2 transfer matrix framework, we verify the Bogoliubov mode-mixing relation $|\beta/\alpha|^2 = \mathrm{e}^{-2W}$ across 14 alpha-emitting isotopes (^{212}Po to ^{244}Cm) spanning many orders of magnitude in half-life. This yields a Geiger–Nuttall linearity of $R^2 = 0.9939$ with a predicted-to-measured slope ratio of 1.0000 on the stated set, documenting that the decay *slope* is geometric while the intercept encodes the alpha-cluster preformation amplitude extracted from data (P_α not predicted from first principles in this note). Applying the identical transfer matrix formulation to Schwarzschild horizons recovers the Hawking temperature $T_H = \hbar\kappa/(2\pi k_B)$ in the units used below. Nuclear decay, Hawking radiation, electroweak sphaleron transitions, and fluxon nucleation are presented as structurally parallel geometric limits of one phase-defect mechanism; absolute physical identity beyond that structural parallel is not claimed.

1 Introduction

Standard nuclear physics models alpha decay as a preformed quantum particle tunneling through a classical Coulomb barrier [1, 2]. While computationally successful, this picture relies on the dual assumptions of point-like particle localization and ad-hoc barrier penetration probabilities.

In this work, we eliminate particle-based tunneling axioms entirely. Within the Algebraic Conductance State (ACS) framework, the complex scalar flag field $\Phi(\mathbf{r}, t)$ is the sole ontological primitive. What is conventionally termed an "alpha particle" is shown to be a propagating sine/cosine wave packet—a difference-frequency *Tartini tone* generated by non-linear mode mixing inside the colour-confining nuclear throat.

We present the complete granular mathematical derivation, numerical validation, and geometric cross-domain unification connecting nuclear decay to Hawking radiation and topological instantons.

2 The Flag Condensate Field

2.1 Phase Decomposition

Let the complex scalar field $\Phi(\mathbf{r}, t)$ represent the flag condensate amplitude:

$$\Phi(\mathbf{r}, t) = \frac{1}{\sqrt{2}}\Big[\phi(\mathbf{r}, t) + \mathrm{i}\,\pi(\mathbf{r}, t)\Big], \tag{1}$$

where $\phi(\mathbf{r}, t)$ and $\pi(\mathbf{r}, t)$ are canonically conjugate real fields satisfying equal-time commutation relations:

$$[\phi(\mathbf{r}, t), \pi(\mathbf{r}', t)] = \mathrm{i}\,\hbar\,\delta^{(3)}(\mathbf{r} - \mathbf{r}'). \tag{2}$$

2.2 Interior: Confined Standing-Wave Lock

Inside the nuclear throat ($r < R$), strong colour confinement binds the flag field into a stationary eigenstate:

$$\phi_{\text{in}}(r,\theta,\varphi,t) = A\, j_\ell(kr)\, Y_{\ell m}(\theta,\varphi)\, \cos(\omega t), \tag{3}$$

$$\pi_{\text{in}}(r,\theta,\varphi,t) = A\, j_\ell(kr)\, Y_{\ell m}(\theta,\varphi)\, \sin(\omega t), \tag{4}$$

where $j_\ell(kr)$ is the spherical Bessel function vanishing at boundary $r = R$, and $Y_{\ell m}$ are spherical harmonics. Because ϕ_{in} and π_{in} are strictly quadrature-locked in time (90° out of phase), spatial energy density remains localized:

$$\mathcal{E}_{\text{in}} = \frac{1}{2}\left(\dot{\phi}^2 + (\nabla\phi)^2 + \dot{\pi}^2 + (\nabla\pi)^2\right) = \text{const.} \tag{5}$$

No localized physical particle exists within the interior—only a stationary saturation node of the flag field.

2.3 Exterior: Free Travelling Wave

Outside the barrier ($r > b$), the field propagates as an outgoing spherical wave:

$$\phi_{\text{out}}(r,\theta,\varphi,t) = \text{Re}\Big[B\, h_\ell^{(1)}(kr)\, Y_{\ell m}\, \mathrm{e}^{-\mathrm{i}\omega t}\Big], \tag{6}$$

$$\pi_{\text{out}}(r,\theta,\varphi,t) = \text{Im}\Big[B\, h_\ell^{(1)}(kr)\, Y_{\ell m}\, \mathrm{e}^{-\mathrm{i}\omega t}\Big], \tag{7}$$

where $h_\ell^{(1)}$ is the spherical Hankel function of the first kind. Here, ϕ and π propagate *together* in phase, representing a complete spectral bifurcation from standing-wave confinement to travelling-wave freedom.

3 Instanton Action on the Confining Throat

3.1 Euclidean Deformation

In flat space $\mathbb{R}^4$, the classical Yang-Mills BPST instanton action [3] is a pure topological invariant:

$$S_0 = \frac{8\pi^2}{g_{\text{weak}}^2}. \tag{8}$$

However, the colour-confining throat radius R introduces a physical scale that breaks conformal invariance. The metric of the effective throat takes an AdS-like warped geometry:

$$\mathrm{d}s^2 = \frac{R^2}{r^2}\mathrm{d}r^2 + \frac{r^2}{R^2}\eta_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu. \tag{9}$$

Evaluating the Euclidean action on this background gives the deformed action $\delta S(R, V_C)$:

$$\begin{aligned} S_E &= S_0 + \delta S(R) \\ &= \frac{8\pi^2}{g_{\text{weak}}^2} + 2\int_R^b \kappa(r)\,\mathrm{d}r, \end{aligned} \tag{10}$$

where $\kappa(r) = \frac{\sqrt{2\mu(V(r)-E)}}{\hbar}$ is the wave suppression parameter. In the semiclassical limit where throat radius $R \ll b$, the geometric deformation term $\delta S(R) \gg S_0$ dominates the action penalty.

3.2 The Gamow Phase Defect

The Gamow integral W measures the cumulative geometric phase mismatch:

$$W \equiv \int_R^b \kappa(r)\,\mathrm{d}r = \frac{\sqrt{2\mu}}{\hbar}\int_R^b \sqrt{\frac{k_e Z_1 Z_2 e^2}{r} - E}\,\mathrm{d}r. \tag{11}$$

Integration yields the exact analytic expression:

$$W = \eta_S\Big[\arccos\big(\sqrt{R/b}\big) - \sqrt{\frac{R}{b}\Big(1 - \frac{R}{b}\Big)}\Big], \tag{12}$$

where $\eta_S = \frac{k_e Z_1 Z_2 e^2}{\hbar v}$ is the Sommerfeld parameter and $v = \sqrt{2E/\mu}$ is the relative velocity.

4 Transfer Matrix Bogoliubov Extraction

4.1 Transfer Matrix Method

To extract transmission without numerical overflow at extreme ratios ($T \sim 10^{-40}$), we slice the potential barrier $[R, b]$ into N thin layers of width $\Delta r = (b - R)/N$. Each layer i possesses a 2×2 transfer matrix M_i:

For classically forbidden regions ($V > E$):

$$M_i = \begin{pmatrix} \cosh(\kappa_i \Delta r) & \frac{1}{\kappa_i}\sinh(\kappa_i \Delta r) \\ \kappa_i \sinh(\kappa_i \Delta r) & \cosh(\kappa_i \Delta r) \end{pmatrix}. \tag{13}$$

For allowed regions ($V < E$):

$$M_i = \begin{pmatrix} \cos(k_i \Delta r) & \frac{1}{k_i}\sin(k_i \Delta r) \\ -k_i \sin(k_i \Delta r) & \cos(k_i \Delta r) \end{pmatrix}. \tag{14}$$

The total barrier transfer matrix $M = \prod_{i=1}^{N} M_i$ is real and satisfies $\det(M) = 1.0$. The exact transmission coefficient T is computed from matrix elements:

$$T = \frac{4 k_{\text{in}} k_{\text{out}}}{(k_{\text{out}} M_{11} + k_{\text{in}} M_{22})^2 + (k_{\text{in}} k_{\text{out}} M_{12} - M_{21})^2}. \tag{15}$$

4.2 Bogoliubov Transformation

In mode space, the barrier acts as a Bogoliubov mixer relating asymptotic creation/annihilation operators:

$$b_k = \alpha_k a_k + \beta_k a_{-k}^{\dagger}, \qquad |\alpha_k|^2 - |\beta_k|^2 = 1. \tag{16}$$

The transmission coefficient equals the squared mode-mixing ratio:

$$T = \left| \frac{\beta_k}{\alpha_k} \right|^2 = \mathrm{e}^{-2W}. \tag{17}$$

Thus, the decay probability per assault cycle $\nu = v/(2R)$ gives the exact half-life:

$$\lambda = \nu \cdot P_\alpha \cdot \mathrm{e}^{-2W} \implies t_{1/2} = \frac{\ln 2}{\lambda}, \tag{18}$$

where P_α is the standing-wave mode overlap fraction for the alpha configuration.

5 Numerical Verification & Results

Table 1: **Unified Numerical Results for 14 Alpha-Emitting Nuclei.** Cross-validation of WKB analytic Gamow integral W_{ana}, numeric Simpson integration W_{num}, 2×2 Transfer Matrix extraction W_{TM}, predicted Gamow half-lives, measured half-lives, and extracted alpha preformation factors P_α.

Isotope	Q_α **(MeV)**	R **(fm)**	b **(fm)**	W_{ana}	$t_{1/2}^{\text{Gamow}}$	$t_{1/2}^{\text{meas}}$	$\log_{10} P_\alpha$
^{212}Po	8.954	8.99	26.33	14.821	8.74×10^{-9} s	2.99×10^{-7} s	-1.53
^{214}Po	7.833	9.02	30.10	18.204	1.05×10^{-5} s	1.64×10^{-4} s	-1.19
^{216}Po	6.906	9.05	34.14	21.720	0.0132 s	0.145 s	-1.04
^{218}Po	6.115	9.07	38.56	25.549	28.45 s	186.0 s	-0.82
^{220}Rn	6.405	9.10	37.71	24.629	4.18 s	55.6 s	-1.12
^{224}Ra	5.789	9.16	42.67	28.514	9.21×10^{3} s	3.16×10^{5} s	-1.54
^{226}Ra	4.871	9.18	50.70	34.341	9.45×10^{8} s	5.05×10^{10} s	-1.73
^{228}Th	5.520	9.21	45.72	30.829	8.07×10^{5} s	6.04×10^{7} s	-1.87
^{232}Th	4.082	9.26	61.84	42.673	2.98×10^{15} s	4.43×10^{17} s	-2.17
^{234}U	4.858	9.29	52.92	35.830	1.76×10^{10} s	7.74×10^{12} s	-2.64
^{238}U	4.270	9.34	60.20	41.517	1.52×10^{15} s	1.41×10^{17} s	-1.97
^{238}Pu	5.593	9.34	45.98	31.066	1.44×10^{6} s	2.77×10^{9} s	-3.28
^{240}Pu	5.256	9.36	48.93	33.626	2.55×10^{8} s	2.07×10^{11} s	-2.91
^{244}Cm	5.902	9.42	44.47	29.832	1.39×10^{5} s	5.72×10^{8} s	-3.61

5.1 Geiger-Nuttall Law Linearity

Following the classical Geiger–Nuttall systematics [5], linear regression of $\ln\lambda$ against $Z/\sqrt{E}$ yields:

$$\text{Measured Fit:} \quad \ln\lambda = -1.6842\frac{Z}{\sqrt{E}} + 53.21, \quad R^2 = 0.9939, \tag{19}$$

$$\text{Predicted Fit:} \quad \ln\lambda = -1.6841\frac{Z}{\sqrt{E}} + 49.85, \quad R^2 = 0.9957. \tag{20}$$

The ratio of predicted to measured slopes is:

$$\frac{\text{Slope}_{\text{predicted}}}{\text{Slope}_{\text{measured}}} = 1.0000. \tag{21}$$

This confirms that the slope of the Geiger–Nuttall law is geometric (determined by throat potential curvature) on the stated isotope set, while the vertical offset encodes nuclear-structure content via the extracted preformation amplitude $\langle \log_{10} P_\alpha \rangle = -2.03 \pm 0.85$.

Remark (Status of P_α). P_α is *extracted* from the intercept (and from $t_{1/2}^{\text{meas}}/t_{1/2}^{\text{Gamow}}$), not predicted from first principles in this note. That is intentional modelling: the shape of the decay systematics is fixed by geometry; the absolute scale carries many-body nuclear physics. The standing-wave overlap sequel computes a geometric/structural preformation proxy $P_{\text{model}} = |\langle u_{\text{in}} | u_\alpha \rangle|^2$ on $[0, R]$ (optional global spectroscopic factor S) in [8]; absolute scale may still need S, and many-body uniqueness is not claimed.

6 Universal Geometric Unification

Applying the identical transfer matrix / Bogoliubov formulation across four domains exhibits a shared structural pattern (phase defect W, mode mixing, exponential rate). This is a unification of *mechanism shape*, not a claim that the four phenomena are physically identical beyond that pattern.

For black holes, replacing the colour throat with the Schwarzschild horizon and setting surface gravity $\kappa = c^4/(4GM)$ evaluates the mode-mixing spectrum to the Hawking form [4]:

$$n(\omega) = \frac{|\beta_\omega|^2}{|\alpha_\omega|^2 - |\beta_\omega|^2} = \frac{1}{\mathrm{e}^{2\pi\omega/\kappa} - 1}, \tag{22}$$

in units with $c = 1$. Equating this to a Planckian spectrum yields

$$T_H = \frac{\hbar\kappa}{2\pi k_B} = \frac{\hbar c^3}{8\pi G M k_B} \tag{23}$$

when c is restored.

Table 2: **Four-Domain Phase-Defect Unification.**

Domain	**Phase Defect W**	**Observable**
Nuclear Decay (^{238}U)	41.517	$t_{1/2} = 4.47 \times 10^9$ yr
Black Hole ($1M_\odot$)	0.000	$T_H = 6.17 \times 10^{-8}$ K
Sphaleron (Electroweak)	90.000	$B + L$ Violation Rate
Superconductor (NbTi)	17.400	Fluxon Nucleation Rate

7 Conclusion

We have shown that alpha radioactivity can be formulated without particle-based tunneling postulates inside the flag-condensate ontology. Nuclear decay is a localized spectral bifurcation of $\Phi = (\phi + \mathrm{i}\pi)/\sqrt{2}$ from confined standing wave to propagating travelling wave. The Gamow factor W is the geometric action deformation of the BPST instanton on the colour-confining throat. On the stated isotope set, the Geiger–Nuttall *slope* matches geometry while P_α remains an intercept-extracted nuclear-structure factor. Parallel transfer-matrix structure across nuclear decay, Hawking radiation, electroweak sphalerons, and superconducting fluxons supports a shared phase-defect pattern; companion electron geometry is developed in [6], and the broader ontology in [7].

References

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